Presented by Nurmukhammed Polat

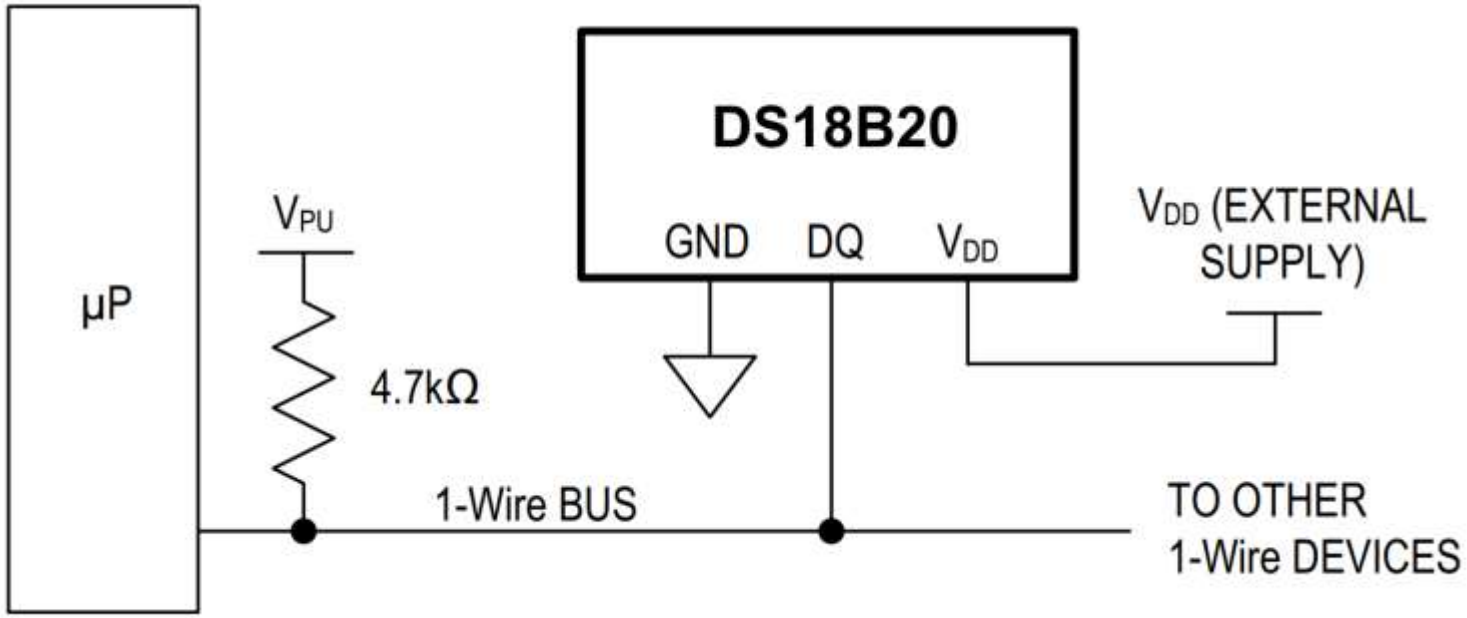
# 1-WIRE TEMPERATURE SENSOR WITH ALERT FUNCTION

# 1-WIRE DS18B20

1 wire protocol is half-duplex serial bus.

Uses 1 data line, 1 power and 1 ground. Only 3 lines needed to function

DS18B20 – the most common temperature sensor. Digital output.



# REQUIRED/BOUGHT HARDWARE AND SOFTWARE

Arduino Uno R4
OLED display (128x32)
DS18B20
Passive Buzzer

Arduino IDE
Libraries:
OneWire
DallasTemperature
Wire
AdafruitGFX
Adafruit_SSD1306

# CONTRUCTED CIRCUIT

# CIRCUIT ON BREADBOARD NOT SOLDERED

**1st Part**

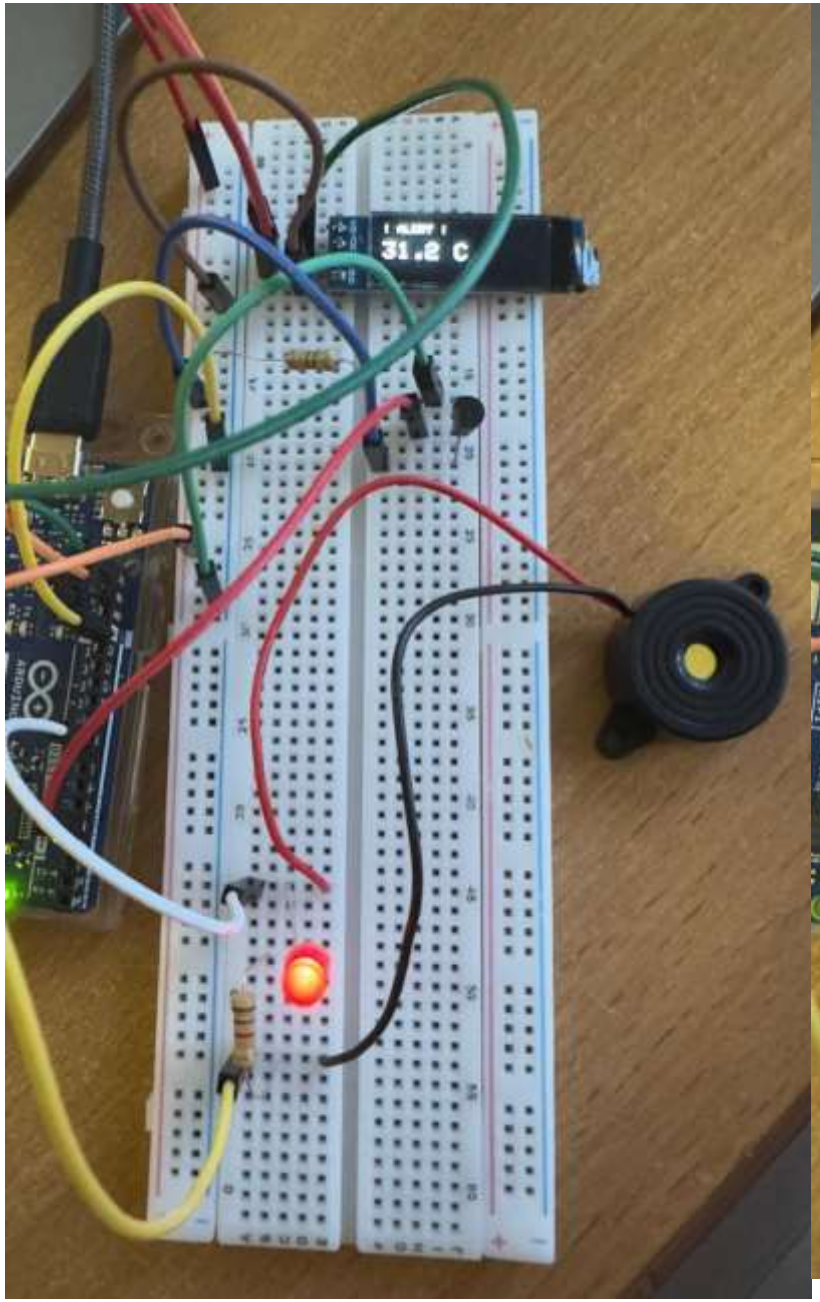
The OLED display on the top which shows the temperaturę is I2C protocol.

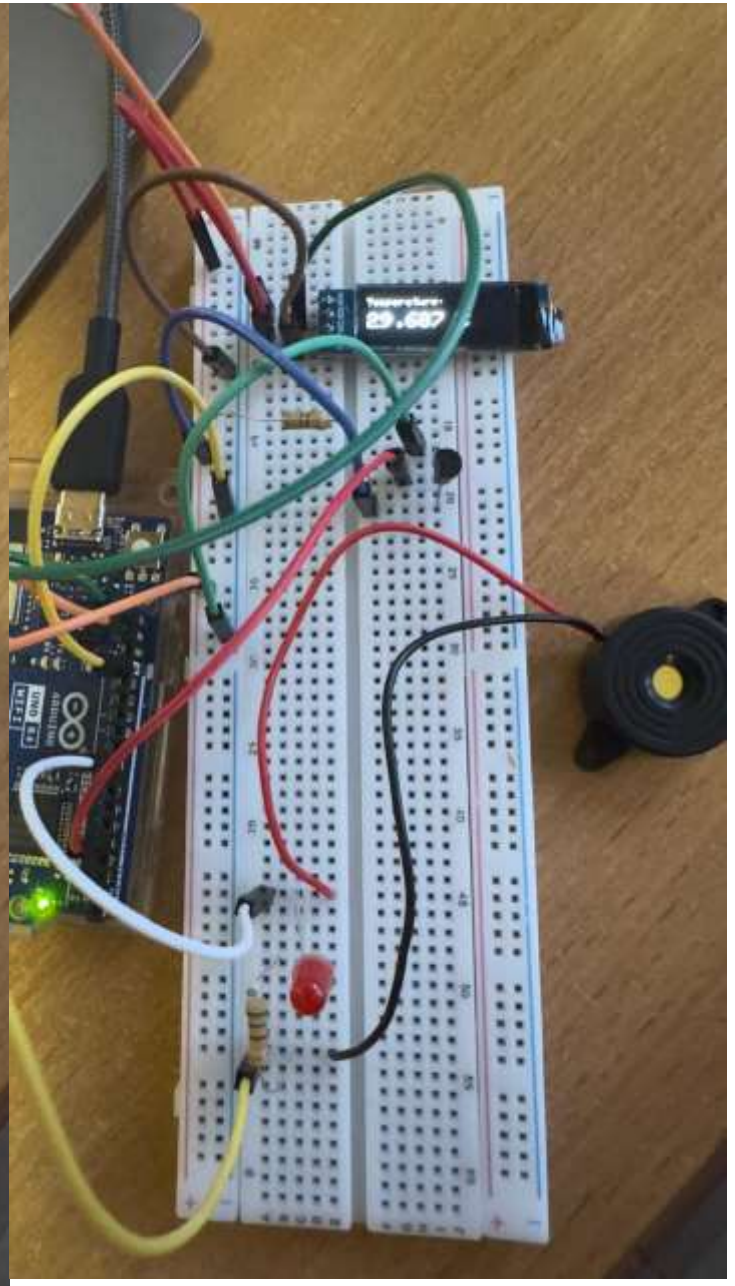
**2nd part**

Under the display temp sensor with pull up resistor(4.7kOhms).

**3rd part**

Buzzer. Positive to pin 8, negative to ground.



# PROBLEMS

# CODE SNIPPET

```
  if (tempC > temp_alert) {
     tone(buzzer, 100);
   }
else {
     noTone(buzzer);
   }

```

The biggest problem was the not soldered microcontroller pins.

Tried changing:

Resistor, Breadboard, jumper wires, sensors

# CONCLUSIONS

SO, THE SENSOR DS18B20 WHICH USES 1-WIRE PROTOCOL WAS USED ALONG WITH OLED DISPLAY AND BUZZER TO ACHIEVE THE OBJECTIVES OF THE PROJECT.